

复习

- 行波法: 特征线法
- 对称延拓法 (热方程也可)
- 分离变量法 (x 2)
- 幂级数法

• Fourier 变换解方程

{ 微分
方程
Parseval 等式
乘号

不一定要算出变换后的函数

题目 1 (15 分)

求如下 Sturm-Liouville 问题的特征值和特征函数

→ 使方程有非零解的 λ

方程的解

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \frac{\lambda}{r^2} R = 0, \quad 1 < r < e,$$

边界条件为

$$R|_{r=1} = 0, \quad \left. \frac{dR}{dr} \right|_{r=e} = 0.$$

$$\text{令 } t = \ln r, \quad (\text{即 } r = e^t), \quad y(t) = R(e^t) = R(r)$$

$$\frac{dy}{dt} = \frac{dR(e^t)}{dt} = e^t \cdot \frac{dR}{dr} = r \cdot \frac{dR}{dr}$$

$$\frac{d^2 y}{dt^2} = e^t \cdot \frac{dR}{dr} + e^t \cdot \frac{d^2 R}{dr^2} \cdot e^t = r \cdot \frac{dR}{dr} + r^2 \cdot \frac{d^2 R}{dr^2}$$

$$\text{原方程: } \frac{1}{r} \left(\frac{dR}{dr} + r \frac{d^2 R}{dr^2} \right) + \frac{\lambda}{r^2} R = 0, \quad r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + \lambda R = 0$$

$$\text{所以 } \Rightarrow \frac{d^2 y}{dt^2} + \lambda y = 0, \quad (0 < t < 1)$$

$$\text{且 } y|_{t=0} = 0, \quad \left. \frac{dy}{dt} \right|_{t=1} = 0.$$

$$\text{通解: } y_n = C_1 \cos \sqrt{\lambda_n} t + C_2 \sin \sqrt{\lambda_n} t$$

$$\text{代入定解条件得 } C_1 = 0, \quad \cos \sqrt{\lambda_n} = 0, \quad \sqrt{\lambda_n} = \frac{2n+1}{2} \pi \quad (n=0, 1, 2, \dots)$$

$$\text{特征值: } \lambda_n = \frac{(2n+1)^2 \pi^2}{4}, \quad n=0, 1, 2, \dots$$

$$y_n = \sin \frac{(2n+1)\pi t}{2}, \quad n=0, 1, 2, \dots$$

$$\text{特征函数 } R_n(t) = \sin \left(\frac{(2n+1)\pi}{2} t \right), \quad n=0, 1, 2, \dots$$

题目 2 (15 分)

试用幂级数解法讨论：当 λ 是何值时一阶积分-微分方程

$$\frac{dy}{dx} - 2xy + \lambda \int_0^x y(t) dt = 0$$

在 $x=0$ 的邻域内存在多项式形式的解？并求出这些解。

$$\text{设 } y = \sum_{n=0}^{\infty} a_n x^n, \quad \frac{dy}{dx} = \sum_{n=0}^{\infty} n a_n x^{n-1}, \quad \int_0^x y(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

$$\text{即 } \sum_{n=0}^{\infty} n a_n x^{n-1} - \sum_{n=0}^{\infty} 2 a_n x^{n+1} + \lambda \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} = 0 \text{ 有解.}$$

$$a_1 + \sum_{n=0}^{\infty} \left[(n+2) a_{n+2} - 2 a_n + \frac{\lambda a_n}{n+1} \right] x^{n+1} = 0.$$

$$\therefore a_1 = 0, \quad (n+2) a_{n+2} = \left(2 - \frac{\lambda}{n+1} \right) a_n.$$

$$a_{2k+1} = 0, \quad k=0, 1, 2, \dots$$

$$a_{2k+2} = \frac{2(2k+1) - \lambda}{(2k+1)(2k+2)} a_{2k}, \quad k=0, 1, 2, \dots$$

要使原方程存在多项式形式的解，则要使有限个 a_{2k} 非零。

$$\text{则取 } \lambda = 4m+2, \quad m=0, 1, 2, \dots, \quad a_{2m+2} = a_{2m+4} = \dots = 0.$$

$$\text{再取 } a_0 = 1, \text{ 则 } a_{2k+2} = \frac{1}{k!} \frac{4(j-m)}{(2j+1)(2j+2)}, \quad a_{2k} = \frac{1}{j!} \frac{4(j-m-1)}{(2j-1) \cdot 2j}, \quad k=1, \dots, m.$$

$$\text{多项式解 } y_m(x) = \sum_{k=0}^m a_{2k} x^{2k}$$

题目 3 (20 分)

(1) 设 $w(x, t)$ 是热方程 $w_t - w_{xx} = 0$ 的解, 证明 $u(x, t) = -2w_x/w$ 满足

$$u_t + uu_x - u_{xx} = 0.$$

(2) 求解下列定解问题:

$$\begin{cases} u_t + uu_x - u_{xx} = 0, & 0 < x < \pi, t > 0, \\ u(0, t) = 0, \quad u(\pi, t) = 0, & t \geq 0, \\ u(x, 0) = \frac{4 \sin 2x}{2 + \cos 2x}, & 0 \leq x \leq \pi. \end{cases}$$

(1) 证明: $u = -2 \frac{w_x}{w}$,

$$u_t = -2 \frac{w_{xt}w - w_x w_t}{w^2},$$

$$u_x = -2 \frac{w_{xx}w - w_x \cdot w_x}{w^2} = -2 \frac{w_t}{w} + 2 \left(\frac{w_x}{w} \right)^2$$

$$u_{xx} = -2 \cdot \frac{w_{xt} \cdot w - w_t w_x}{w^2} + 2 \cdot 2 \cdot \frac{w_x}{w} \cdot \left(-\frac{1}{2} u_x \right)$$

$$= u_t - 2 \frac{w_x}{w} \cdot u_x = u_t + u \cdot u_x \neq$$

(2) 设 $w(x, t)$ 满足 $\begin{cases} w_t - w_{xx} = 0, & 0 < x < \pi, t > 0 \\ w(x, 0) = 2 + \cos 2x, & 0 \leq x \leq \pi \\ w_x(0, t) = w_x(\pi, t) = 0, & t \geq 0 \end{cases}$

解: 原方程组的解 $u(x, t) = -2 \frac{w_x}{w} = \frac{4 \sin 2x \cdot e^{-4t}}{2 + \cos 2x \cdot e^{-4t}}$

设 $w(x, t) = \sum_{n=0}^{\infty} X_n(x) T_n(t) = \frac{4 \sin 2x}{2e^{4t} + \cos 2x}$

$$\begin{cases} X_n''(x) + \lambda_n X_n = 0 \\ X_n'(0) = X_n'(\pi) = 0 \end{cases} \quad \begin{cases} X_n = C_1 \cos \sqrt{\lambda_n} x + C_2 \sin \sqrt{\lambda_n} x \\ C_2 = 0, \quad \sin \sqrt{\lambda_n} \pi = 0, \quad \sqrt{\lambda_n} = n \end{cases}$$

$$\therefore \lambda_n = n^2, \quad X_n(x) = \cos n x \quad n = 0, 1, 2, \dots$$

$$\sum_{m=1}^{\infty} X_m T_m' + \sum_{m=1}^{\infty} \lambda_m X_m T_m = 0, \quad T_n' + \lambda_n T_n = 0, \quad T_n = C_n e^{-n^2 t}$$

$$\therefore w(x, t) = \sum_{n=0}^{\infty} \cos n x \cdot C_n e^{-n^2 t}, \quad w(x, 0) = \sum_{n=0}^{\infty} C_n \cos n x = 2 + \cos 2x$$

$$\therefore C_0 = 2, \quad C_2 = 1, \quad C_n = 0 \quad (n=1, 3, 4, \dots), \quad w(x, t) = 2 + \cos 2x \cdot e^{-4t}$$

题目 B2 (15 分)

定义函数 $f(t)$, $-\infty < t < +\infty$ 的 Fourier 变换为

$$\mathcal{F}[f(t)](\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt.$$

(1) 求证: $\mathcal{F}[f(t+t_0)](\omega) = e^{i\omega t_0} \mathcal{F}[f(t)](\omega)$;

(2) 利用 Fourier 变换求解一阶偏微分方程初值问题

$$\begin{cases} u_x + u_y = f(x, y), & x \geq 0, -\infty < y < +\infty, \\ u(0, y) = \varphi(y), & -\infty < y < +\infty. \end{cases}$$

(2) 关于 y 作 Fourier 变换

就算 φ 给出了显式函数也不要填入 $[f], [f_y]$ 的具体表达式

$$\begin{aligned} \text{(1) 证明: } \mathcal{F}[f(t+t_0)](\omega) &= \int_{-\infty}^{+\infty} f(t+t_0) e^{-i\omega t} dt \\ &= \int_{-\infty}^{+\infty} f(t+t_0) e^{-i\omega(t+t_0)} \cdot e^{i\omega t_0} dt \\ &= e^{i\omega t_0} \int_{-\infty}^{+\infty} f(t+t_0) e^{-i\omega(t+t_0)} dt \\ &= e^{i\omega t_0} \mathcal{F}[f(t)](\omega). \end{aligned}$$

(2) 解: 关于 y 作 Fourier 变换,

$$\mathcal{F}_y[u_x](\omega) = \frac{d}{dx} \hat{u}(\omega)$$

$$\mathcal{F}_y[u_y](\omega) = (i\omega) \cdot \hat{u}(\omega)$$

$$\frac{d}{dx} \hat{u} + i\omega \cdot \hat{u} = \hat{f}, \quad \hat{u}(x, \omega) = C \cdot e^{-i\omega x} + \int_0^x \hat{f}(\xi, \omega) e^{-i\omega(x-\xi)} d\xi$$

$$\hat{u}(0, \omega) = C = \hat{\varphi}(\omega), \quad \hat{u}(x, \omega) = \hat{\varphi}(\omega) \cdot e^{-i\omega x} + \int_0^x \hat{f}(\xi, \omega) e^{-i\omega(x-\xi)} d\xi$$

$$\begin{aligned} u &= \mathcal{F}^{-1}[\hat{u}(x, \omega)](y) = \mathcal{F}^{-1}[\hat{\varphi}(\omega) \cdot e^{-i\omega x}] + \mathcal{F}^{-1}\left[\int_0^x \hat{f}(\xi, \omega) e^{-i\omega(x-\xi)} d\xi\right] \\ &= \varphi(y-x) + \int_0^x f(\xi, y-x+\xi) d\xi. \end{aligned}$$

题目 4 (15 分)

(1) 已知函数 e^{-x^2} 的 Fourier 变换是

$$\mathcal{F}[e^{-x^2}](\lambda) = \int_{-\infty}^{+\infty} e^{-x^2} e^{-i\lambda x} dx = \sqrt{\pi} e^{-\frac{1}{4}\lambda^2}.$$

设 $a > 0$ 为常数. 试求函数 $e^{-a\lambda^2}$ 的 Fourier 逆变换:

$$\mathcal{F}^{-1}[e^{-a\lambda^2}](x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-a\lambda^2} e^{i\lambda x} d\lambda.$$

(2) 用 Fourier 变换法求解初值问题:

$$\begin{cases} u_t - u_{xx} - 2tu = 0, & -\infty < x < +\infty, t > 0, \\ u(x, 0) = \varphi(x), & -\infty < x < +\infty, \end{cases}$$

的 Green 函数 $G(x, t, \xi)$; 即求函数 $G(x, t, \xi)$ 使得

$$u(x, t) = \int_{-\infty}^{+\infty} G(x, t, \xi) \varphi(\xi) d\xi.$$

$$\begin{aligned} (1) \quad \mathcal{F}^{-1}[e^{-a\lambda^2}](x) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-a\lambda^2} e^{i\lambda x} d\lambda \\ &= \left[\frac{1}{2\pi} \int_{-\infty}^{+\infty} \sqrt{\pi} e^{-\frac{(2\sqrt{a}\lambda)^2}{4}} \cdot e^{i \cdot \frac{x}{2\sqrt{a}} \cdot (2\sqrt{a}\lambda)} d(2\sqrt{a}\lambda) \right] \cdot \frac{1}{2\sqrt{a}\pi} \\ &= e^{-\left(\frac{x}{2\sqrt{a}}\right)^2} \cdot \frac{1}{2\sqrt{a}\pi} = \frac{1}{2\sqrt{a}\pi} e^{-\frac{x^2}{4a}}. \end{aligned}$$

(2) 关于 x 作 Fourier 变换: $\mathcal{F}_x[u(x, t)](\lambda) = \hat{u}(\lambda, t)$

$$\mathcal{F}_x[u_t](\lambda) = \frac{d}{dt} \hat{u}(\lambda, t)$$

$$\mathcal{F}_x[u_{xx}](\lambda) = (i\lambda)^2 \hat{u}(\lambda, t) = -\lambda^2 \hat{u}(\lambda, t)$$

$$\begin{cases} \frac{d}{dt} \hat{u}(\lambda, t) + (\lambda^2 - 2t) \hat{u}(\lambda, t) = 0 \\ \hat{u}(\lambda, 0) = \hat{\varphi}(\lambda) \end{cases}$$

$$\Rightarrow \hat{u}(\lambda, t) = \hat{\varphi}(\lambda) e^{-\lambda^2 t + t^2}$$

$$\mathcal{F}^{-1}[e^{-\lambda^2 t + t^2}](x) = e^{t^2} \cdot \frac{1}{2\sqrt{\pi t}} \cdot e^{-\frac{x^2}{4t}} = \frac{1}{2\sqrt{\pi t}} e^{-\frac{x^2}{4t} + t^2} = G(x, t, \xi)$$

$$u(x, t) = \varphi(x) * \left(\frac{1}{2\sqrt{\pi t}} e^{-\frac{x^2}{4t} + t^2} \right) = \int_{-\infty}^{+\infty} \varphi(\xi) \cdot \boxed{\frac{e^{-\frac{(x-\xi)^2}{4t} + t^2}}{2\sqrt{\pi t}}} d\xi.$$